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标题	Ring-closing metathesis of prochiral oxaenediynes to racemic 4-alkenyl-2-alkynyl-3,6-dihydro-2H-pyrans
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摘要（原文节选）

The prochiral 4-(allyloxy)hepta-1,6-diynes, optionally modified in the positions 1 and 7 with an alkyl or ester group, undergo a chemoselective ring-closing enyne metathesis yielding racemic 4-alkenyl-2-alkynyl-3,6-dihydro-2H-pyrans. Among the catalysts tested, Grubbs 1st generation precatalyst in the presence of ethene (Mori conditions) gave superior results compared to the more stable Grubbs or Hoveyda – Grubbs 2nd generation precatalysts. This is probably caused by a suppression of the subsequent side-reactions of ⋯

数据表 · Table 1（7 行，含数值行 5 行）

Optimization of the catalyst loading depending on compound 2 triple bond substitution (NMR yields)a.

substituent	cat. loading (%)	yield (%)
H	1	0
H	5	92
H	10	>98
Me	1	>98
Me	5	>98

脚注（逐字）：aG-I catalyst was used.

正文对应句（逐字）：Room temperature (25 ° C) was chosen as it is common for this type of reactions. In selected cases, we attempted to rise the temperature to improve the conversion but without success. Moreover, the high volatility of most of the substrates resulted in a loss of material unless sealed systems were use

数据表 · Table 2（13 行，含数值行 6 行）

RCEYM of oxaenediynes 2 (NMR yields).

substrate	R	product	catalyst					
G-I	G-II	HG-II						
atmosphere	atmosphere	atmosphere						
ethene	argon	ethene	argon	ethene	argon			
yield (%)	yield (%)	yield (%)						
2a	H	12a	92a	>98	35b,c	7	<5b,c	<5b
2b	Me	12b	>98a	78	28b,c	61	40b,c	64
2c	Et	12c	>98a	53	29b,c	66	27b,c	70
2d	Pr	12d	>98a	52	13b,c	98	67b,c	90
2e	CO2Me	12e	>98a	>98	18b,c	36	50b,c	78
2f	SiMe3	12f	20c	0	<5b,c	0.5	<5b,c	11

脚注（逐字）：aIsolated yields: 12a: 75%, 12b: 98%, 12c: 87%,12d: 95%, and 12e: 65%; bcontains side-products from CM with ethene (Figure 3); cthe yield was estimated from the inseparable product mixture by careful analysis of the vinylic area in the NMR spectra.

正文对应句（逐字）：Room temperature (25 ° C) was chosen as it is common for this type of reactions. In selected cases, we attempted to rise the temperature to improve the conversion but without success. Moreover, the high volatility of most of the substrates resulted in a loss of material unless sealed systems were use

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